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Topic: Weather Forecasting using Python

BRANCH:CSE(AIML)

SCHOOL:SCAI

SUBJECT:Fundamentals in AIML

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PROJECT REPORT

1) Weather Reporter Application Using Python:

The Weather Reporter application functions as a command-line program which uses Python to access current weather data through an application programming interface for any city which users specify. Users can enter the name of a city and access weather information such as temperature, humidity, wind speed, and weather conditions.

The project demonstrates my ability to create Python applications which use API systems to exchange data and handle JSON data through user-friendly interface design. The demonstration will show how contemporary software programs obtain their operational data through external information sources which help them achieve more effective information gathering.

2. Introduction:

People currently use information to determine their decisions throughout modern society. People use weather information to create predictions which help them decide about various activities. The Weather Reporter Application provides real-time weather information through its command-line interface for any location which users request.

The application functions through OpenWeatherMap API which delivers worldwide weather data. Through this API I will create a Python command-line application which enables users to retrieve and display weather information.

The project enhances understanding in these areas:

* API communication

* Data parsing

* Exception handling

* User interaction in Python

3. Problem Statement:

Many users require quick access to weather information without navigating complex applications or websites. People need a lightweight system that delivers immediate weather updates to them.

The project addresses the problem by delivering two main features:

* Users receive current weather information

* Users can access the system through a straightforward interface

* The system operates without requiring resource-intensive software

4. Objectives:

The primary goals of this project work toward achieving two main objectives.

* The team will create a weather reporting application using Python programming.

* The team will add external API support to their Python software through implementation work.

* The team will develop methods for effective JSON data management.

* The team will develop procedures to manage system errors.

* The team will design a command-line interface that users will find easy to use.

* The team will learn about how HTTP requests work together with HTTP responses.

5. Scope of the Project:

The Weather Reporter Application project will include the following functions.

<u>Current Scope:</u>

* The system will retrieve actual weather conditions from the Internet.

* The system will show temperature and humidity and wind speed together with weather conditions.

* The system will enable users to make multiple search requests.

Future Scope:

* The system will develop a graphical user interface application.

* The system will develop a mobile application that connects with their existing software.

* The system will create a voice recognition system for users to ask about weather information.

* The system will enable users to interact with the system in different languages.

* The system will store users' previous search results for later access without Internet connection.

6. Literature Review:

The historical development of weather applications shows major transformation throughout its existence. People used television and newspapers together with their weather forecast systems during earlier times. The modern world uses APIs to deliver current weather information which applications can access in real time.

Modern applications use:

* REST APIs

* Cloud-based data services

* Real-time data processing

The project applies the same basic concepts from previous work but uses a simpler educational system that teaches Python programming skills.

7. System Requirements

Hardware Requirements:

* Computer/Laptop

* Minimum 4GB RAM

* Internet connection

Software Requirements:

* Python 3.x

* Requests library

* Code editor (VS Code / PyCharm)

8) Algorithm

The project employs a structured methodology which consists of three distinct steps.

Step 1: User Input

The user enters the name of a city.

Step 2: API Request

The application sends a request to the OpenWeatherMap API.

Step 3: Data Retrieval

The API delivers weather information to the application through JSON format.

Step 4: Data Processing

The application extracts relevant data fields.

Step 5: Display Output

The extracted data is presented in a format that users can easily comprehend.

Step 6: Loop Execution

The system continues its functions until the user decides to stop.

9. System Architecture:

Components:

1. User Interface (CLI)

2. Python Application Logic

3. API Server (OpenWeatherMap)

4. Data Processing Module

Flow:

User → Python Program → API Request → API Response → Data Processing → Output

10. Algorithm:

The program begins with its execution through its starting point which leads to a display of a welcome message. The system needs to receive city name input from the user. The program stops when the user enters exit as their input. The system sends an API request to the server. The system receives a JSON response from the server. The system checks the response status to determine its outcome. The system processes successful requests by performing the following tasks.

11. Implementation:

The system retrieves weather information from the database which displays the results. The system displays an error message to the user. The process continues until further notice. The code divides into three separate sections which include Function definition and User interaction loop and API handling The application uses HTTP GET requests to fetch weather data. The system uses JSON responses to extract temperature and humidity and weather description and wind speed data. The system handles three types of errors which include network errors and invalid inputs and API failures.

12. Code Explanation:

The requests module operates as a tool which enables users to perform HTTP requests. The get_weather function handles API requests and shows the obtained data. The URL contains The URL contains The URL contains The URL contains The URL contains The URL contains City name API key Units metric The response handling section first verifies the response status through Http code checking Which handles two cases 200 code indicates success while all other codes represent an error situation. The system operates continuously until the user chooses to stop it.

13. Output Description:

The output section describes the results which will be produced through the system. The system will display its current weather information for Delhi The system output will display Delhi weather information which includes Temperature°30 Condition°Clear sky Humidity°45 Wind Speed°3 m/s. The system output will display Delhi weather information which includes Temperature°30 Condition°Clear sky Humidity°45 Wind Speed°3 m/s.

14. Advantages:

* The system provides users with an interface which requires no advanced skills to operate.

* The system delivers current information to users at all times.

* The system functions as a small resource application.

* The system provides users with an interface which requires no advanced skills to operate.

* The system allows users to create new functions through existing components.

15. Disadvantages:

* The system needs an active internet connection to function properly.

* The system needs an application programming interface to function.

* The system provides users with a restricted set of functions.

* The system does not provide any visual presentation elements.

16. Applications:

* People use this system to check their local weather conditions.

* The system serves as a learning resource for educational institutions.

* The system functions as an application programming interface learning resource.

* The system provides users with command-line interface tools.

17. Future Enhancements

The upcoming project will implement five main enhancements which include a Tkinter-based GUI a mobile application weather forecasting automatic location detection via GPS and voice assistant compatibility.

The testing section includes all test cases which are described below.

18. Testing:

Test Cases:

Test Case	Input	Expected Output
Valid city	London	Weather data
Invalid city	xyz123	Error message
Empty input	""	Prompt again
Exit	exit	Program ends

19. Conclusion :

The Weather Reporter Application demonstrates Python capabilities through its ability to access external APIs which provide current weather information. The project demonstrates API integration and JSON parsing and user interaction through its simple design.

The project provides a basic platform which enables future advanced application development while offering developers hands-on experience with actual programming tasks.

20. BIBLIOGRAPHY

* OpenWeatherMap CONCEPTS

* Python AUTHENTICATION

* Requests API DOCUMENTATION RELATION

* Websites ON HELP AND TRICKS